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Research Paper

Estimating cumulative input effects in annual crop production: A LASSO-based panel data approach from India

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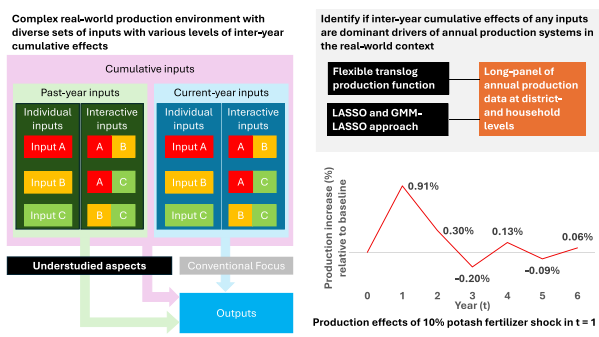
HIGHLIGHTS

- Cumulative effects of agricultural inputs identified under farmer field conditions.
- LASSO and GMM-LASSO applied to district- and farm-level panel data from India.
- Cumulative quantity of potassium remains key input, including in Rigorous-LASSO.
- Potassium application in one year tends to reduce its usage in the following year.
- One-time potassium shock affects productivity and production for multiple years.

GRAPHICAL ABSTRACT



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ABSTRACT

CONTEXT: In non-experimental settings, evidence of interannual cumulative effects of inputs for annual crops (e.g., the effects of inputs in one year on outputs in the subsequent year) remains limited despite potential implications within dynamic production systems. The scarcity of sufficiently long annual panel data partly explains this. Furthermore, in non-experimental settings, quantities of many inputs are highly correlated with one another and across years, posing challenges when isolating such cumulative effects through conventional estimation methods.

OBJECTIVE: This study narrows these knowledge gaps by applying novel methods to unique annual panel datasets at district- and farm household-levels in India. Specifically, we identify whether certain inputs exhibit meaningfully significant cumulative effects on production relative to the non-cumulative effects and the effects of other inputs.

METHODS: We start with flexible translog production functions appropriate for identifying cumulative effects in non-experimental settings. We apply shrinkage methods (LASSO and GMM-LASSO) to approximate production functions with reduced parameter dimensions, addressing multicollinearity among multiple inputs as well as among the same inputs across years, and potential endogeneity in inputs.

RESULTS: Throughout the shrinkage process, potassium fertilizer consistently remains a key predictor of outputs, while other inputs (land, labor, capital, irrigation, and other fertilizer nutrients) drop out mainly due to high collinearity with potassium and other inputs. More importantly, the cumulative quantity of potassium from the

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previous year, as well as the current year, is a consistently more critical determinant of production than the quantity of potassium from the current year alone, demonstrating the significant cumulative effects of potassium. These patterns hold both at district and farm levels across diverse agroecologies and cropping systems. Furthermore, the dynamic panel data analyses further suggest that farmers' use of potassium in the current year is significantly negatively affected by its use in the previous year, potentially stabilizing outputs across years. Earlier agronomic results suggesting residual effects of potassium are potentially relevant across wider geographic regions than previously thought. Simulation exercises reveal that the cumulative effects of potassium translate into a significant carryover of productivity into the following year and, combined with the dynamics, considerable repercussions on production during the subsequent years

SIGNIFICANCE: To the authors' knowledge, this is the first study to verify the interannual cumulative effects of potassium fertilizer in nonexperimental settings, using methods that control for all other relevant production inputs and their potential cumulative effects within the flexible translog production function framework

1. Introduction

The cumulative effects of agricultural inputs on production have been recognized as an integral driver of production systems dynamics (e.g., Henryson et al., 2018; Jernigan et al., 2020; Rui et al., 2020; Aboah and Enahoro, 2022; Jones et al., 2025). The effects of inputs in one year on the system productivity in subsequent years have often been considered to determine the strength and overall significance of the intertemporal, dynamic linkages within the production systems. For example, farming experiences acquired from previous years often affected labor productivity in subsequent years (e.g., Berger, 2001). Using agricultural equipment has often affected soil quality in subsequent years (e.g., Lyu et al., 2019), while irrigation has affected hydrological conditions which could alter the land quality for subsequent years (e.g., Singh, 2020). Among others, inorganic fertilizer has been considered another potentially important input that could have cumulative effects on production not only in the current year but also in the subsequent year through the buildup of soil nutrients and lagged effects (e.g., Meena et al., 2019).

Nonetheless, there has been limited real-world, field-level evidence on the cumulative effects of inputs carried over from the previous year and their significance on production (both overall level and annual fluctuations) within the annual crop system. The scarcity of such evidence can lead to suboptimal recommendations for input use by extension/advisory and misguided public support programs for specific inputs in a given year. The issue has been particularly relevant in developing countries where the cumulative effects of, for example, fertilizer, could be further affected by insufficient soil fertility management knowledge/techniques in maximizing benefit from residual soil nutrient (e.g., Stevens, 2018), inadequate institutional capacity for related advisory services (Sheikh et al., 2024), and/or credit constraints that could lead to complex interannual dynamics in input use decisions (e.g., Magazzino et al., 2024). Potential cumulative effects of fertilizer nutrients have been suggested anecdotally or through agronomic studies (e.g., Johnston, 1988; Askegaard et al., 2004; Harper et al., 2012; Kihara et al., 2016; Meena et al., 2019; Xia et al., 2023; Rotili et al., 2023), many of which suggesting potentially significant cumulative effects of various fertilizer nutrients. For example, under experimental settings for cotton production in the US, as much as 70 % or more of potassium K was carried over to the next production season (Harper et al., 2012). In the US, in a separate experimental setting, continuous application of liquid dairy manure over 12 years has led to 30–50 % increases in soil phosphorus and potassium contents after the 12 years (Rui et al., 2020). In an experimental setting in Argentina, balanced application of fertilizer nutrients, including increased sulfur (S) and phosphorus (P), led to 10–13 % higher yield for soybean even 6 years later (Rotili et al., 2023). However, they have less commonly been examined at representative field levels that reflect farmers' actual production conditions. Addressing this knowledge gap is critical because whether cumulative effects are significant at the field level affects optimal application rates of these inputs in given years (e.g., Harper et al., 2012). Flexible representation of production relations, including translog production function, has been among the best ways to characterize technologies underlying complex production systems and have been increasingly used in agronomic studies (e.g., Lu et al., 2018; Asante et al., 2019; Van Dijk et al., 2020; Chen et al., 2024). However, addressing the aforementioned knowledge gap within the field-level production function framework has remained challenging due to various reasons. Long panel data on agricultural production required for such analyses has been generally scarce. Estimation has been complicated due to multicollinearity between specific fertilizer nutrients with other fertilizer nutrients (e.g., among N, P and K) as well as with other inputs, and multicollinearity between inputs use in current year and previous years, not to mention potential endogeneity of input variables.

We narrow this knowledge gap using annual panel data of agricultural production at both district levels (the District Level Database for Indian Agriculture and Allied Sectors (*DLD-data* hereafter) (ICRISAT and TCI, 2024)) and household levels (the Village Dynamics in South Asia (VDSA) project of the International Crops Research Institute for the Semi-Arid Tropics (ICRISAT) (ICRISAT, 2024) (*VDSA-data* hereafter)) from India, which provides unique evidence of the lagged/cumulative effects of fertilizer. We characterize production system with a flexible translog production function, and apply classes of Least Absolute Shrinkage and Selection Operator (LASSO) methods (Tibshirani, 1996; Caner, 2009a; Shi, 2016a; Ahrens et al., 2020) to address multicollinearity issues among input variables and robustly compare the current-year effects and lagged effects of nitrogen, phosphate, and potassium, together with all the other inputs. Furthermore, we apply dynamic panel regressions (Arellano and Bond, 1991; Wooldridge, 2005) to identify inter-annual dynamic relations of fertilizer use which illustrates the extent to which their cumulative effects translate into interannual fluctuations in their use. Lastly, we conduct simulation exercises to understand the implications of cumulative effects and input use dynamics on production system dynamics.

Our empirical approaches constitute novel contributions. First, we obtain critical evidence based on unique panel data with annual frequency and sufficient length, which are particularly suitable for identifying the cumulative effects of inputs in the real-world field-level production contexts. Our data also includes district-level data for which endogeneity is of less concern, the type of data that is less commonly used for most developing country contexts. Second, applying LASSO to a flexible translog production function enables the identification of dominant inputs and associated cumulative effects (among a broad range of inputs and their possible combinations, as well as the timing of the use of these inputs), so that the identified cumulative effects are of major relevance within the production system. Our LASSO-based selection approach significantly improves on the translog production function literature, which often resorted to rather ad hoc approaches in

reducing parameter dimensions (e.g., Fan and Pardey, 1997). Our approach of incorporating multi-year input use also relaxes restrictive assumptions commonly used in standard specifications of the translog production function, in which only inputs from the same year as outputs are typically included. Third, our analysis on inter-annual dynamic relations of input use based on the aforementioned dynamic panel regression is, to the best of our knowledge, the first application to the specific case of fertilizer, including potassium, in developing countries. The estimates from these aforementioned novel methods are then used as key inputs in the subsequent simulation exercise to demonstrate the quantitative significance of the identified cumulative effects and associated dynamics of input use, in terms of size and duration, specifically, how a shock in potassium use or price has long-lasting production and productivity shocks for subsequent years.

Although our results highlight the cumulative effects of potassium, our analytical approach also considers the whole set of inputs, such as land, labor, capital, and other expenses. Therefore, our results are not because we focus specifically on potassium but rather because potassium turns out to be the input for which cumulative effects matter (unlike those for other inputs) in India. In other words, our results show that cumulative effects are significant only for potassium, and similar cumulative effects are somewhat limited for other inputs and other fertilizer nutrients. Analyzing the cumulative effects within the production framework, rather than reduced form fertilizer response framework, also avoids us from erroneously concluding about cumulative effects of potassium, when in fact the true effects may be occurring through other inputs that are highly correlated with potassium.

Our results suggest that the aggregate quantity of potassium from the current year as well as the previous year, is a consistently more important determinant of production than the quantity of potassium from the current year alone, and quantity of all the other inputs. Potassium use is strongly affected by its dynamic relationship with the previous year's. Cumulative effects of potassium translate into a significant carryover of productivity to the following year and, combined with the dynamics, considerable repercussions on production during the subsequent years.

India is a suitable case for our analyses. Primary nutrients (N, P, K) have become increasingly critical inputs in India's agricultural production. In 2022, India accounted for approximately 20 % of nitrogen, phosphate, and 5 % of potassium used for agriculture globally (FAO, 2024). Furthermore, India decontrolled prices of Phosphatic and Potassic (P and K) fertilizers in 2011, potentially leading to significant spatiotemporal variations in the quantity of fertilizer used, which, combined with the LASSO-approach, can be exploited to identify their effects on production.

Our study also makes general contributions to various strands of literature. We contribute to the literature on production system dynamics in general, and specifically to the literature on production function with cumulative effects of inputs, such as in the production of perennial crops (e.g., Chand, 1994), roles of infrastructures and Research & Development (R&D) (Fan and Pardey, 1997), or natural resource stock (e.g., Fuglie et al., 2021), by offering an empirical approach to distinguish the effects of accumulated inputs from non-accumulated inputs. We also contribute to the studies that highlight the accumulated or lagged effects of certain fertilizer nutrients (e.g., Johnson, 1986; Askegaard et al., 2004; Kihara et al., 2016) by providing formal evidence at field levels. Lastly, methodologically, our study contributes to the growing literature on model selections (Tibshirani, 1996) and their applications to the agricultural contexts (e.g., Mayer et al., 2019; Asadollah et al., 2024; De Clercq and Mahdi, 2024; Sunoj et al., 2024) through the application of LASSO-approach to robustly compare the predictive power of the cumulative inputs with single same-year inputs alone.

The remainder of this paper is structured as follows. Section 2 describes our data and descriptive statistics. Section 3 discusses regression-based analyses of production function and potassium use dynamics. Section 4 presents the simulation exercises that illustrate key quantitative implications of the estimated production function structures. Section 5 discusses key overall implications on inter-annual dynamic linkages of agricultural production systems. Finally, section 6 concludes.

2. Materials and descriptive statistics

2.1. Materials used

DLD-data are district-level historical data compiled by the Tata-Cornel Institute for Agriculture and Nutrition (TCI) and the ICRISAT. These district-level data have been compiled by the TCI and the ICRISAT historically from various organizations within India, including Directorate of Economics and Statistics, Directorate of Agriculture, Agriculture department, State Agricultural Marketing Board, Directorate of Veterinary Services, State Level Bankers Committee, National Bank for Agriculture and Rural Development, Public Works Department, Directorate of Animal Husbandry, Horticulture and Food Processing Department, Directorate of Horticulture, Department of Child and Women Development, Department of Environment &Forest (ICRISAT and TCI, 2024).

DLD-data capture annual agricultural production, production values, area cultivated, agricultural workers, agricultural credit used for the purchased services and inputs (such as machines, agrochemicals), the share of land irrigated, and fertilizer use (including nitrogen, phosphate, and potash) aggregated at the district level for the period between 2002 and 2016. We focus on 496 districts for which all production and inputs data are available (approximately 85 % of all districts). We also drop approximately 5 % of year-district observations with missing information on some inputs. We thus use unbalanced panel samples of 7054 district-level observations from 496 districts from 2002 through 2016. Fig. 1 illustrates the 496 districts included in our study.

VDSA-data are farm-household level data compiled by the ICRISAT. The sampling frame of VDSA-data consists of a total of 18 purposively selected villages in 5 Southern and Western Indian states (Andhra Pradesh, Gujarat, Karnataka, Madhya Pradesh, and Maharashtra). Through the census of households in each village, households were classified into four groups according to farm size and land ownership (large = owning 1.5 acre or more total farm land in the initial survey round; medium = owning 1–1.5 acre; small = owning less than 1 acre; land less = do not own farm land), from which 10 households (40 households total in each village) were randomly selected and interviewed annually between 2010 and 2014. After dropping observations with missing information, we focus on the 3041 panel observations in the VDSA-data.

In addition, historical rainfall data and drought data are obtained from Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS) (Funk et al., 2015) and Standardized Precipitation Evapotranspiration Index (SPEI) (Vicente-Serrano et al., 2010), respectively. Nighttime luminosity data are obtained from NOAA (2022). Soil property data are obtained from FAO (2012). For each survey year, these data are extracted as averages for each district and for each village location in VDSA data (soil data are time-invariant and extracted as constant across years).

2.2. Descriptive statistics

Table 1 and Table 2 summarize the descriptive statistics of variables used from DLD-data and VDSA-data, respectively. During the 2002–2016 period, the district typically produced 7.3 million Rupee of outputs annually for all the major commodities combined, cultivated

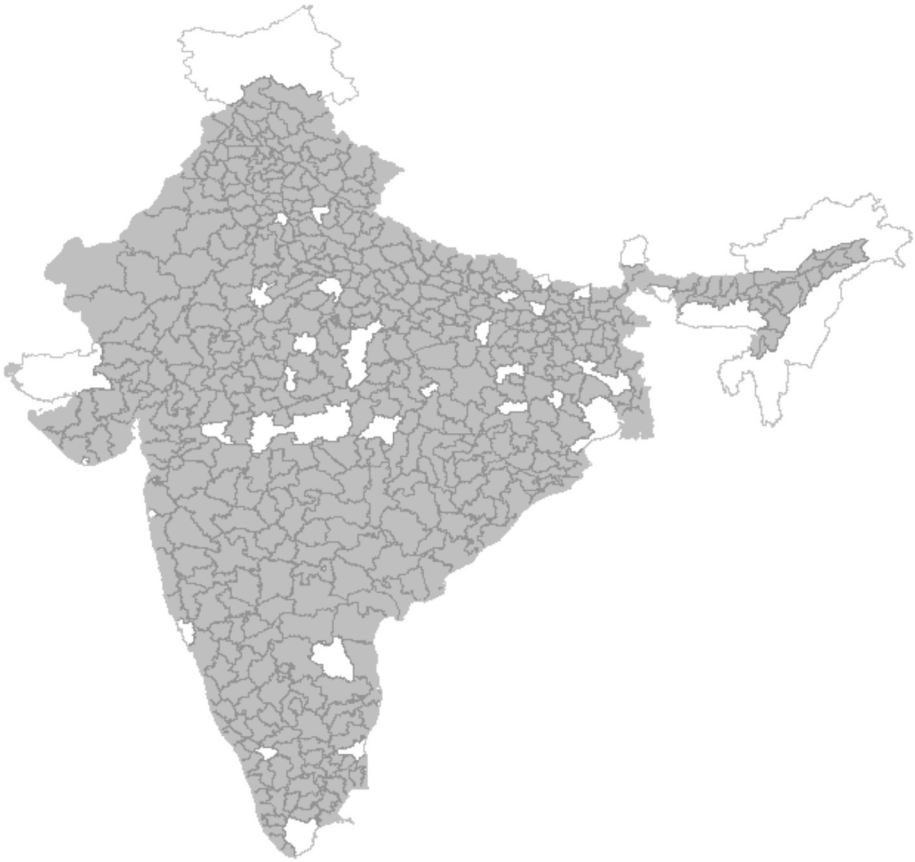

Fig. 1. Districts included in the analyses with DLD-data (shaded districts).
Source: Authors.

Table 1
Descriptive statistics of DLD-data.

Variables	Mean	Std.dev
Production revenue (current million Rupee)	7.268	8.804
Land (1000 ha)	261.090	213.434
Agricultural population (1000)	483.367	305.931
Agricultural credit (current million Rupee)	6.644	11.059
Share of land irrigated	0.489	0.317
Nitrogen used (1000 ton)	27.555	26.467
Phosphorus used (1000 ton)	11.165	11.646
Potassium used (1000 ton)	4.538	7.047
Nighttime light (district average, nanoWatt per cm^2 per $radian^2$)	5.543	7.202
Absolute rainfall deviations (absolute value of z-score)	0.793	0.569
Absolute drought index deviations (absolute value of z-score)	0.806	0.609
Number of sample districts	496	
Total sample size	7054	

Source: Authors' compilation from DLD-data.

about 261,000 ha of land, about half of which were irrigated (Table 1). About 6.6 million Rupee of agricultural credit was taken out for the purchase of various services and inputs (mostly machines, mechanization services, agrochemicals). District level use of nitrogen, phosphorus, and potassium are on average 27.6, 11.2, and 4.5 million tons annually. There are significant variations across districts, with standard deviations in the same order of magnitude as mean (i.e., coefficient of variation around 1).

Typical farm households in VDSA data produced 84,000 and 34,000 Rupee of agricultural products annually, at mean and median, respectively, during 2010–2014 period (Table 2). Most are smallholders cultivating 1–2 ha in total, using 300–400 person-hours of labor, and using purchased inputs (other than fertilizer) worth 20,000–40,000 Rupees. About 30 % used irrigation. The use of machines in terms of horsepower-hours is 257 and 12 hp-hours at mean and median, respectively. A typical household owned 1.25 ha of land, with household size of 5, and owned agricultural equipment worth 8000 Rupees at median. They face wages and mechanization service fees of around 21 and 450 rupees per hour, respectively. As is consistent with nationwide trends (Kishore et al., 2021), the price of nitrogen had remained generally stable around 6–8 Rupee/kg, prices of phosphorous and potassium increased from 11 to 25 Rupee/kg, and from 5 to 6 to 18 Rupee/kg, respectively.

3. Empirical methods

Our overall empirical approach consists of three sequential steps (Fig. 2 illustrates the overall analytical flow of our paper), with later steps partly guided by the earlier steps. In step 1, we conduct production function analyses through the LASSO-process to assess whether any production inputs exhibit significant cumulative effects that are dominant within the production system. As is shown subsequently, potassium is identified as one of the production inputs that exhibit such effects. We then proceed to assess the implications on such cumulative effects in actual production outcomes. Doing so requires examining how the use of potassium is determined by farmers, and in particular, how it is dynamically affected by the potassium use in the previous year. For this purpose, in step 2, we conduct another econometric analysis that

Table 2
Descriptive statistics of VDSA-data (annual total).

Variables	Mean	Median	Std.dev
Production revenue (current 1,000 Rupee)	83.914	34.436	168.895
Land cultivated (ha)	2.104	1.113	3.54
Labor use (person-hours)	408.176	281	444.823
Use of own machines (horsepower-hour)	257.495	12	1041.677
Purchased inputs excl. fertilizer (1,000 Rupee)	37.106	18.181	62.716
Use irrigation (yes = 1)	0.299	0	0.458
Nitrogen used (kg)	98.604	39	196.314
Phosphorus used (kg)	67.943	25.76	135.792
Potassium used (kg)	12.013	0	38.154
Land owned (ha)	3.196	1.25	5.771
Household size	5.53	5	2.65
Agricultural equipment owned (1000 rupees)	48.674	7.8	176.073
Wage (current rupee per hour)	22.252	21.124	9.486
Mechanization service fees (rupees per hour)	451.566	450	127.625
Price – Nitrogen (current rupee per kg)	7.472	7	2.402
2010	6.438	6.25	
2011	8.055	7	
2012	7.913	7.527	
2013	7.467	7.28	
2014	7.445	7.125	
Price – Phosphorus (current rupee per kg)	20.905	23.5	7.823
2010	11.737	11	
2011	16.502	16.35	
2012	25.062	23.5	
2013	25.281	25	
2014	25.781	25	
Price – Potassium (current rupee per kg)	13.535	14.5	5.282
2010	6.653	5.25	
2011	8.796	8.5	
2012	15.809	14	
2013	18.394	18	
2014	17.938	17.508	
Total sample size	3,104		

Source: Authors' compilation from VDSA-data.

assesses the inter-annual dynamics in potassium use. Lastly, in step 3, using the relevant parameter estimates from steps 1 and 2, we conduct simulation exercises to identify dynamic productivity and production effects of a shock in potassium use, to illustrate the quantitative implications of the estimated cumulative effects.

3.1. Regression based characterization of intertemporal linkages within production systems

3.1.1. Estimating flexible translog production function through shrinkage regression

For both DLD-data and VDSA-data, we start with a translog production function,

$$\ln Y_{ht} = \alpha + \sum_X \beta_X \ln X_{ht} + \frac{1}{2}\sum_X \sum_{X'} \beta_{XX} \ln X_{ht} \ln X^*_{ht} + \beta_S S_{ht} + f_h + \varepsilon_{ht}$$
$$(h = \text{district } j \text{ for DLD data, farm household } i \text{ for VDSA data}) \quad (1)$$

Y_{ht} is the total crop production value in year t. X_{ht} is a vector of inputs including land, labor, capital (proxied by agricultural credit taken) and irrigation. X_{ht} also includes quantity of each of three fertilizer nutrients, nitrogen (N_{ht}), phosphate (P_{ht}) and potassium (K_{ht}). X^*_{ht} is alias to X_{ht}. S_{ht} is a vector of other exogenous time-variant variables. f_h is time-invariant fixed effects for panel h. α and β's are estimated parameters, while ε_{ht} is an idiosyncratic error term.

For DLD-data, control variables S_{jt} include district-average nighttime light index (proxy for the level of overall economic development), as well as weather shocks proxied by absolute z-scores of annual total rainfall and annual average drought index (SPEI) relative to their historical distributions during the pre-sample period (1980–2000). S_{jt} also includes the year dummy and its interactive terms with state dummy variables. For VDSA-data, S_{it} includes the year dummy and its interaction with village dummy variable to account for any year-specific village-level shocks.

Eq. (1) is a standard translog production function without explicit cumulative effects of inputs. In our empirical analyses, we expand (1) by

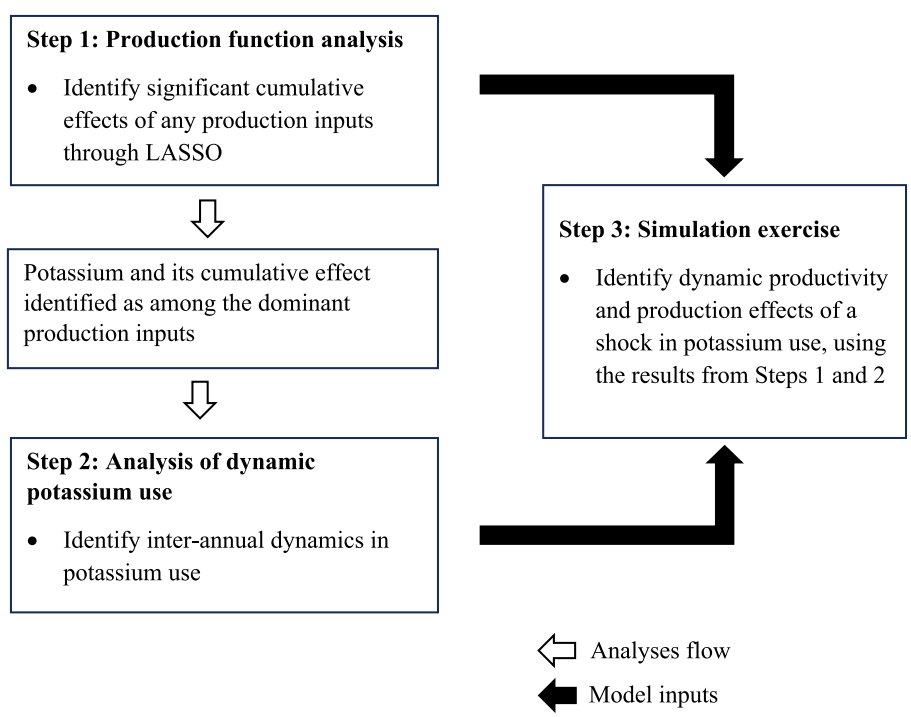

Fig. 2. Overview of analytical flow.
Source: Authors.

adding lagged input variables, so that

$$\begin{aligned}\ln Y_{ht} = \alpha &+ \sum_X \beta_{X1} \ln X_{ht} + \frac{1}{2}\sum_X \sum_{X^*} \beta_{X11} \ln X_{ht} \ln X^*_{ht} + \sum_X \beta_{X0} \ln X_{h,t-1} \\ &+ \frac{1}{2}\sum_X \sum_{X^*} \beta_{X00} \ln X_{h,t-1} \ln X^*_{h,t-1} + \frac{1}{2}\sum_X \sum_{X^*} \beta_{X10} \ln X_{ht} \ln X_{h,t-1} \\ &+ \sum_X \beta_{XC} \ln X_{htc} + \frac{1}{2}\sum_X \sum_{X^*} \beta_{XC1} \ln X_{ht} \ln X_{htc} \\ &+ \frac{1}{2}\sum_X \sum_{X^*} \beta_{XC0} \ln X_{h,t-1} \ln X_{htc} + \frac{1}{2}\sum_X \sum_{X^*} \beta_{XCC} \ln X_{htc} \ln X^*_{htc} \\ &+ \beta_S S_{ht} + f_h + \varepsilon_{ht} \\ &(h = \text{district } j \text{ for DLD data, farm household } I \text{ for VDSA data})\end{aligned} \tag{2}$$

in which

$$X_{htc} = \frac{X_{ht} + X_{h,t-1}}{2}, \tag{3}$$

with "c" denoting "cumulative".

We define that an input has cumulative effects if both β_{X1} and β_{X0}, or β_{XC}, is statistically significantly different from 0. The functional form (2) serves as an initial specification for the LASSO-process to narrow down most influential variables among X_{ht}, $X_{h,t-1}$ and X_{htc}. Conceptually, X_{htc}, $X_{ht}X_{htc}$, $X_{h,t-1}X_{htc}$ are redundant since they are subsumed in X_{ht}, $X_{h,t-1}$ and $X_{ht}X_{h,t-1}$. Nonetheless, X_{htc}, $X_{ht}X_{htc}$, $X_{h,t-1}X_{htc}$ are included for empirical practicality, primarily to detect cumulative effects when X_{ht} and $X_{h,t-1}$ are highly correlated. If a certain input exhibits cumulative effect, both X_{ht} and $X_{h,t-1}$ (as well as their interaction terms with other common inputs), have significant coefficients in true production function. However, if there is significant multicollinearity between X_{ht} and $X_{h,t-1}$, the LASSO-process may select only one of X_{ht} or $X_{h,t-1}$, forcing us to incorrectly conclude that cumulative effects are absent. For example, in our case, adding $X_{h,t-1}$ to (1) increases the median Variance Inflation Factor (VIF) across all input variables by almost 100 times compared to the case when only X_{ht} is included. If X_{htc} is also included in the model and approximates well the cumulative use of the input in t and $t-1$, the LASSO-process may select X_{htc} (while dropping X_{ht} nor $X_{h,t-1}$), detecting the evidence of cumulative effects. Fig. 3 provides a simple illustration of how Eq. (2) is aimed to differentiate pathways through which input in t (X_{ht}) and $t-1$ ($X_{h,t-1}$) affect the output in t (Y_{ht}).

While LASSO-process, both standard LASSO for DLD-data and GMM-LASSO for VDSA-data, is a central tool for our estimation, its technical detail diverts our focus on production system analysis. We therefore delegate detailed discussion of the LASSO-process to Appendix A. Importantly, as described in Appendix A, our LASSO-process also accounts for the potential heteroskedasticity in the error term which can be common in the context of agricultural production.

In estimating (2) for DLD-data, we also examined that stationarity of all variables used. For a long panel data like DLD-data, lack of stationarity in output and input variables can lead to spurious regression. Our analyses indicate that all variables in DLD-data are stationary (Appendix B and Table B1 provide more detailed descriptions and stationarity test results).

Importantly, because we focus on LASSO and the translog production function, our primary results are applicable only to the linear approximation of the input-output relations within the production systems, and readers should interpret the results with caution. Nonetheless, the translog production function represents a second-order approximation to an unknown arbitrary function, including a nonlinear function (Denny and Fuss, 1977). Not only is the translog production function suitable for serving as a linear approximation of a complex production system (Denny and Fuss, 1977), but it also encompasses some of the other commonly-used functional forms as special cases (e.g., Cobb-Douglas function) and thus provides broad empirical relevance. Thus, our approach can still capture key aspects of the underlying production systems that involve complex, nonlinear relations. We also focus on LASSO that is suitable for analyzing linear relations, rather than other approaches directly applicable to nonlinear relations (e.g., other Machine Learning approaches including Random Forest, Neural Network, among others) because our primary objective lies in the interpretation of the production system (i.e., the contributions of each input (together with its cumulative effect)), rather than prediction or forecasting of outputs. LASSO enables us to interpret the estimated production systems more clearly, compared to some of the aforementioned approaches which are more tailored for prediction/forecasting and may be less suitable for model interpretation due to their "black box" nature (e.g., Andrews et al., 1995; May et al., 2011; Haddouchi and Berrado, 2019). Nonetheless, as part of the robustness check, we also provide brief exploratory results based on Neural Network analysis to show that the implications of our primary results are not simply the artifacts of LASSO-based analysis.

3.1.2. *Dynamics of fertilizer nutrients use at farm household level (VDSA-data)*

As is shown later in the results section, analyses of (2) through LASSO-approach suggest significant effects of potassium K_{htc}. We conduct supplementary analyses using VDSA-data to show that the dependence on significant cumulative effects has crucial implications on the use of potassium fertilizer, i.e., dynamic mechanisms determining the use of potassium, which extends beyond static mechanism often assumed for fertilizer in the literature. We also compare the results with

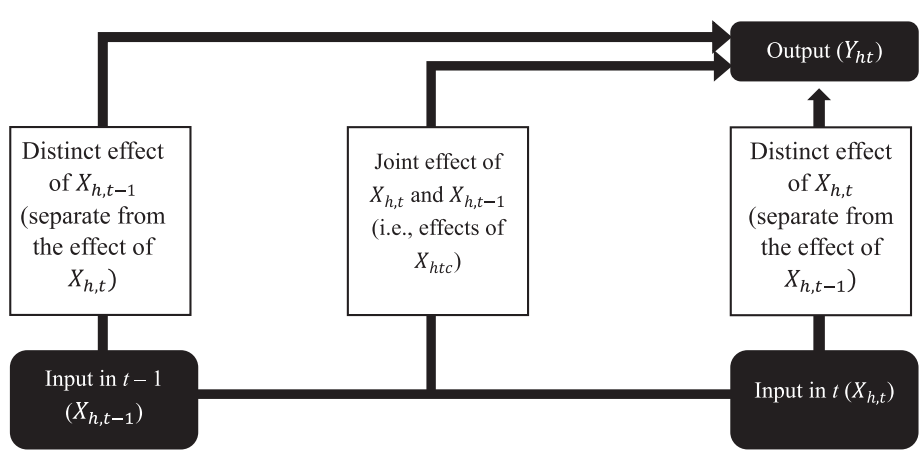

Fig. 3. Potential pathways of the effects of lagged inputs.
Source: Authors.

those of nitrogen which is found as another important variable in the later section but without cumulative effects, to identify patterns unique to potassium.

Given the challenges associated with dynamic panel-data estimation (e.g., biases due to correlation between unobserved fixed-effects and lagged dependent variable under the standard specifications (Nickell, 1981)) and the censured nature of the quantity of fertilizer use at farm levels (i.e., quantity is either positive or censored at 0), suitable models must be carefully selected for the dynamic panel data analysis. Specifically, we assess the interannual dynamics in potassium use (as well as nitrogen use for comparison) using 2 commonly used econometric methods in the literature on dynamic panel data analyses; Arellano and Bond (1991) method which can be used to gain insights into the interannual dynamics of potassium (as well as nitrogen) use at extensive margin, and Wooldridge's (2005) method which can directly model the interannual dynamics for censored variables (intensive margin).

3.1.2.1. Arellano and Bond (1991) Dynamic panel linear probability model. We first assess the dynamics at the extensive margin (i.e., whether to use or not in a particular year). Separate from the dynamics at the intensive margin (as described in the following subsection), the fertilizer use decision at the extensive margin can have its distinctive mechanism in developing countries. For example, a high transactions costs in input markets including those for fertilizer, common in developing countries, can serve as an entry barrier that affect decisions at extensive margins, while such barriers may be less relevant once farmers entered the market and now are making decisions at the intensive margins (e.g., Xu et al., 2009; Takeshima and Liu, 2020). Dynamics at the extensive margin can also be readily estimated in a Linear Probability Model (LPM) framework within the well-established Arellano and Bond (1991) linear dynamic panel data method, as previously done in the literature (e.g., López and Yadav, 2010; Kerr et al., 2014; Takeshima et al., 2023). Specifically, we estimate

$$Q_{it}^{*} = \gamma + \gamma_Q Q_{i,t-1}^{*} + \gamma_{\Pi}\Pi_{it} + \gamma_S S_{it} + \gamma_Z Z_{it} + h_i^{*} + \upsilon_{it} \quad \text{(for each of nitrogen and potassium)} \tag{4}$$

in which Q_{it}^{*} equals 1 if i used fertilizer nutrient in t (for each of nitrogen and potassium) and 0 otherwise. Π_{it} is the price of each nutrient faced by i in t. The parameter h_i^{*} is unobserved farm household fixed effects. As is well-known, $Q_{i,t-1}^{*}$ can be endogenous due to its correlation with compound disturbance term $h_i^{*} + \upsilon_{it}$, a common challenge in dynamic panel model. Arellano and Bond (1991) estimator addresses this endogeneity by instrumenting lagged dependent variable by Π_{it}, S_{it} and the transformation of lagged variables in earlier t.

3.1.2.2. Wooldridge's (2005) dynamic panel Tobit model. We also assess the dynamics at the intensive margins, by applying Wooldridge's (2005) dynamic panel random-effects Tobit model. Specifically, we estimate,

$$Q_{it} = \gamma + \gamma_Q Q_{i,t-1} + \gamma_{Q0} Q_{i,0} + \gamma_{\Pi}\Pi_{it} + \gamma_S S_{it} + \gamma_Z Z_{it} + \widetilde{h}_i + \upsilon_{it} \tag{5}$$

in which $Q_{it}(\geq 0)$ is the quantity of fertilizer nutrients used (each of nitrogen and potassium). Notation $\widetilde{h}_i$ is time-invariant random effects. $Q_{i,0}$ is the function based on the quantity of each type of fertilizer nutrient used in the initial survey years. Notations γ's represent estimated parameters, and υ_{it} represents idiosyncratic errors. Including $Q_{i,0}$ as an explanatory variable provides consistent estimate of the coefficient γ_Q (Wooldridge, 2005). We approximate $Q_{i,0}$ by the average values of Q_{it} over the first 3 years (2010−2012), and focus the analyses on the last 3 years (2012–2014). $Q_{i,0}$ based on average over the 3 initial years (2010–2012) is likely to provide stable and reliable approximation of the initial condition by smoothing out idiosyncratic shock in any single year. Past studies similarly use multiple-years pre-sample means of dependent variable to approximate initial conditions in similar dynamic panel data framework (e.g., Blundell et al., 1999, 2002; Máñez et al., 2020; Sanchis Llopis et al., 2024).

3.2. Productivity and production implications of cumulative effects of potassium: simulation-based assessment

As is shown in the results section (section 4), production function estimations reveal significant roles of cumulative quantity of potassium. We conduct additional simple simulation exercises to gain insight into what such cumulative effects mean in terms of overall productivity and production changes over time. Since the goal of our simulation exercise is to gain insights within the specific framework of the regression analyses presented in the previous section and to focus on aspects directly supported by the associated evidence, we keep the setup of the simulation exercise simple rather than adding any complexities.

Our simulation exercises focus on quantifying the effects of a shock represented by a 10 % increase in potassium uses in the initial period, on the changes in two aspects of the subsequent period; productivity-type effects, and the actual production effects. As is shown subsequently, the first aspect highlights how the cumulative effects of potassium serve the role of *carrying over* productivity from the current year to the next year, as well as the magnitude of such carryover. The second aspect highlights how such dynamic productivity effects, combined with interannual dynamics in potassium use, characterize the nature of the repercussions in subsequent years following the aforementioned initial shock in potassium use, as well as how significant they are in magnitude.

For this simulation exercise, we focus on the production function results based on DLD-data and potassium use dynamics results on VDSA-data, while not considering production function results on VDSA-data. This is because production function results for VDSA-data estimated through GMM-LASSO are more suitable for qualitatively identifying statistically significance effects of potassium in the presence of endogeneity and for showing the relevance of cumulative effects at farm household levels, while less suitable for deriving simulated quantitative impacts.

3.2.1. Productivity effects of an increase in potassium use in the previous year

We first conduct simple simulation exercises to assess how significantly, in terms of magnitude, the production in year t is affected by the potassium use in year $t-1$. To focus on the pure attribution of such production changes solely to the potassium use in the previous year, we keep the use of other inputs in both t and $t-1$ unchanged (including the potassium use in year t (K_{jt})). Since the production change in t that is not attributable to inputs use in t can be synonymous with a change in productivity, we interpret this as *productivity effects* of potassium use from $t-1$. This enables us to simulate the effects directly based on the estimated coefficients of variables that contain K_{jtc} from section 3, so that

$$\Delta\widehat{Y}_{jt} = \frac{\partial Y_{jt}}{\partial K_{j,t-1}} = \frac{\partial Y_{jt}}{\partial K_{jtc}} \cdot \frac{\partial K_{jtc}}{\partial K_{j,t-1}} = \frac{\partial Y_{jt}}{\partial K_{jtc}} \cdot \frac{\partial\left(\frac{K_{jt}+K_{j,t-1}}{2}\right)}{\partial K_{j,t-1}} = \frac{\partial Y_{jt}}{\partial K_{jtc}} \cdot 0.5 \tag{6}$$

in which $\Delta\widehat{Y}_{jt}$ is the changes in production in district j in year t attributable to the change in potassium use in year $t-1$ (ignoring, for now, the effects on the changes in potassium use in t, which will be relaxed in separate simulation model below). The relevant estimated coefficients from production function (2) and (3) are then used to derive the values for the term $\frac{\partial Y_{jt}}{\partial K_{jtc}}$ in (6).

3.2.2. Multi-year production effects of an initial increase in potassium use

We then run another simple simulation to gain insights into how inter-annual productivity changes shown in 3.2.1 translate into actual production changes, when we also consider the changes in potassium

use in $t+1$ as a dynamic responses to potassium use in year t (using the estimated results of Eq. (5)). Specifically, when incorporating the dynamic relations between $K_{j,t-1}$ and $K_{j,t}$, the effects of a marginal change in $K_{j,t-1}$ in actual cumulative amount of potash, K^*_{jtc}, becomes

$$\frac{\partial K^*_{jtc}}{\partial K_{j,t-1}} = \frac{\partial\left(\frac{K^*_{jt}+K_{j,t-1}}{2}\right)}{\partial K_{j,t-1}} = \left(\frac{\partial K^*_{jt}}{\partial K_{j,t-1}} + \frac{\partial K_{j,t-1}}{\partial K_{j,t-1}}\right)\cdot 0.5 = \left(\frac{\partial K^*_{jt}}{\partial K_{j,t-1}} + 1\right)\cdot 0.5.$$

By replacing $\frac{\partial K_{jtc}}{\partial K_{j,t-1}}$ in (6) (which does not account for dynamic changes in potassium use) with $\frac{\partial K^*_{jtc}}{\partial K_{j,t-1}}$ (which accounts for dynamic changes in potassium use), we have

$$\Delta Y^*_{jt} = \frac{\partial Y^*_{jt}}{\partial K_{j,t-1}} = \frac{\partial Y_{jt}}{\partial K_{jtc}} \cdot \frac{\partial K^*_{jtc}}{\partial K_{j,t-1}} = \frac{\partial Y_{jt}}{\partial K_{jtc}} \cdot \left(\frac{\partial K^*_{jt}}{\partial K_{j,t-1}} + 1\right)\cdot 0.5. \quad (7)$$

ΔY^*_{jt} represents changes in production in district j in year t attributable to the change in potassium use in $t-1$ while, unlike $\Delta \widehat{Y}_{jt}$, also accounting for the changes in potassium use in t. Production function coefficients from (2) and (3) are again inserted for $\frac{\partial Y_{jt}}{\partial K_{jtc}}$ as in productivity change simulation 3.2.1, while coefficients for the lagged potassium variable (quantity at $t-1$) from the results for (5) are inserted for the term $\frac{\partial K^*_{jt}}{\partial K_{j,t-1}}$, respectively, in (7).

As described in the previous section, our LASSO estimation also shows that extending lags to $t-2$ and beyond generally does not outperform covering lags only up to $t-1$. These results imply that the cumulative effects of a shock in potassium use in t may primarily hold in $t+1$ and start fading from $t+2$ on. To gain deeper insights into how the effects of an initial shock on potassium use remains (albeit in a diminishing manner) over medium-term, we simulate (7) up to $t+5$ for two scenarios, one with initial shock and the other without any initial shock (baseline scenario). Comparing these two scenarios allow us to estimate effects of an initial shock in potassium in t for the subsequent 5 years.

4. Results

4.1. Regression results

4.1.1. District-level production function based on DLD-data

Table 3 presents the LASSO-selected input variables for (2) and how the set of key variables narrows down as the penalty weight (λ) on the number of variables increases (see Appendix A on the role of λ). A higher λ is associated with a more parsimonious production function, and λ selected by theory-driven ("rigorous") LASSO approaches (RLASSO hereafter) is often considered desirable. RLASSO selects λ based more heavily on statistical theories underlying the LASSO process, compared to other LASSO approaches that rely more on within-data trial-and-error validation processes that are not grounded in theories (Chernozhukov et al., 2016; Ahrens et al., 2020) (a more technical description of RLASSO is provided in Appendix A). We therefore show RLASSO-selected results as our primary focus while also showing less parsimonious results based on smaller λ.

Table 3
Set of LASSO-selected parameters (DLD-data).

LASSO penalization parameter (λ)	400	500	600	690.65 (RLASSO-selected λ)
LASSO selected parameters of translog function[a]	$N_{jt}A_{jt}$ $K_{jtc}A_{jt}$ $A_{jt}W_{jtc}$ $K_{jtc}K_{jtc}$ $A_{jt}A_{jt}$ $K_{jtc}N_{jt}$ $N_{jt}N_{jt}$	$N_{jt}A_{jt}$ $K_{jtc}A_{jt}$ $A_{jt}W_{jtc}$ $K_{jtc}K_{jtc}$ $A_{jt}A_{jt}$ $K_{jtc}N_{jt}$	$N_{jt}A_{jt}$ $K_{jtc}A_{jt}$ $A_{jt}W_{jtc}$ $K_{jtc}K_{jtc}$	$N_{jt}A_{jt}$ $K_{jtc}A_{jt}$ $A_{jt}W_{jtc}$

Source: Authors based on LASSO estimation.
[a] A = land, W = capital, N = nitrogen, K = potassium.

Table 4
Post Double-Selection (PDS) LASSO-approximation of translog production function for district level data (DLD-data) by different levels of penalization parameter λ[a,b].

LASSO-selected variables	$\lambda = 400$	$\lambda = 500$	$\lambda = 600$	$\lambda =$ 690.65 selected by RLASSO	$\lambda =$ 690.65 selected by RLASSO
					Post Double-Selection LASSO
	Coef. (std. err)	Coef. (std. err)	Coef. (std. err)	Coef. (std. err)	Coef. (std. err)
$N_{jt}A_{jt}$	−0.098*** (0.033)	0.004 (0.014)	0.025*** (0.008)	0.025*** (0.008)	0.027 (0.043)
$\boldsymbol{K_{jtc}A_{jt}}$	**0.042*** **(0.024)**	**−0.030*** **(0.017)**	**−0.007** **(0.013)**	**0.024***** **(0.005)**	**0.035***** **(0.008)**
$A_{jt}W_{jtc}$	0.014 (0.010)	0.011 (0.010)	0.022*** (0.008)	0.014** (0.007)	0.008 (0.026)
$\boldsymbol{K_{jtc}K_{jtc}}$	**0.022**** **(0.009)**	**0.012** **(0.008)**	**0.015***** **(0.006)**		
$A_{jt}A_{jt}$	0.098*** (0.029)	0.062** (0.027)			
$\boldsymbol{K_{jtc}N_{jt}}$	**−0.037*** **(0.019)**	**0.017** **(0.012)**			
$N_{jt}N_{jt}$	0.052*** (0.015)				
Nighttime light	Included	Included	Included	Included	Included
Weather shocks	Included	Included	Included	Included	Included
Year * State	Included	Included	Included	Included	Included
District fixed effects	Included	Included	Included	Included	Included
Intercept	Included	Included	Included	Included	Included
Samples	7054	7054	7054	7054	7054
Average returns to scale				0.960	1.038

Source: Authors. Asterisks indicate statistical significance: ***1 % **5 % *10 %.
[a] A = land, W = capital, R = irrigation, N = nitrogen, P = phosphate, K = potassium.
[b] Bold texts refer to coefficients that contain $\boldsymbol{K_{jtc}}$.

Table 4 presents the estimated coefficients of RLASSO-selected input variables, estimated through Post Double Selection LASSO method (Belloni et al., 2014a) which allows causal interpretations of LASSO-selected input variables, as well as simple LASSO-selected regressions for different λ's. At $\lambda = 400$, a total of 7 variables are selected. At $\lambda =$ 690.65 (the RLASSO-selected λ), only 3 key variables remain, $N_{jt}A_{jt}$ $K_{jtc}A_{jt}$, and $A_{jt}W_{jtc}$, in which N, A, K and W are nitrogen, land, potassium, and capital respectively. Importantly, we are interested in which variables are consistently selected across different λ, and in particular, whether variables containing lagged value of potassium K_{jtc} are more consistently selected than those containing only non-lagged value of potassium (K_{jt}). Table 3 shows that variables containing K_{jtc} ($K_{jtc}A_{jt}$, together with $K_{jtc}K_{jtc}$ and $K_{jtc}N_{jt}$) are always selected regardless of the level of parsimony, while variables containing K_{jt} are not. Furthermore, at least one of these variables containing K_{jtc} remain consistently statistically significant, indicating its importance as key predictors of production function (2). Fig. B1 in Appendix B illustrates how these variables ($N_{jt}A_{jt}$, $K_{jtc}A_{jt}$, and $A_{jt}W_{jtc}$, bold lines) remain significant as the

models become more parsimonious through RLASSO-methods and less significant input variables are eliminated. These results suggest that K_{jtc} that includes lagged quantity of potassium in $t-1$ is a consistently better determinant of production than K_{jt} that only includes potassium in t, within the flexible translog production function specification (2) for DLD-data. Put differently, results are consistent with the argument that potassium has more cumulative effects in production that persist across years (at least beyond the current year).

Table 4 bottom row indicates that the sample average returns to scale (computed using Kim's (1992) method) is generally around 1, i.e., constant returns to scale, suggesting that our RLASSO-selected result is a reasonable approximation of a standard production function.

Importantly, our results remain generally unchanged when we further expand the time lags and include input variables from earlier years (i.e., $X_{j,t-2}, X_{j,t-3}, \ldots$) in the model, and also add expanded set of X_{jtc} that also include cumulative quantity of inputs from these earlier years. Our additional analyses (not shown) indicate that these inputs from $t-2$ or earlier are rarely selected in our LASSO-process. These patterns indicate that the cumulative effects of input in t may start fading 2 years later and subsequent years.

Input variables in DLD-data can be reasonably assumed as exogenous to output because of their measurement at district level, rather than at farm household level. Nonetheless, given the non-experimental setting underlying the DLD-data, questions remain regarding whether the production function coefficients of K_{jtc} can be interpreted as more a causal effect rather than simply a correlation. While it is difficult to fully distinguish them in our setting, some indicative evidence can be obtained from a falsification test that is based on the idea that future input use should not affect the current output. Appendix B (B.3) summarizes the results of the falsification test. As is described there, the results are consistent with our hypothesis that the coefficients for variables consisting of K_{jtc} can be reasonably interpreted as causal effects.

4.1.1.1. Robustness across heterogeneous production systems. Table 4 describes the average conditions that hold for the whole of India. We provide further evidence for sub-samples split across major agroecological conditions and cropping systems within India, and replicated the LASSO-estimation to each sub-sample to assess potential heterogeneity.

4.1.1.1.1. Soil types. Table B2 (Appendix B) summarizes the results for subsamples split at median values of key soil properties at district-level, namely, soil organic contents, alkalinity (pH) and drainage levels, particularly important factors affecting the effectiveness of potassium in India (e.g., Wakeel and Ishfaq, 2022). Patterns are generally similar to Table 4, across districts with different soil organic contents and drainage properties, as well as districts with above-median soil alkalinity; variables containing K_{jtc} are more consistently selected than those containing only K_{jt}. While no variable containing K_{jtc} is selected by LASSO in districts with below-median soil alkalinity, neither are variables containing K_{jt}; in other words, evidence does not contradict Table 4 that K_{jt} is not any more influential than K_{jtc}.

4.1.1.1.2. Cropping systems. Table B3 summarizes sub-sample results based on average cropping systems during the study period, i.e., grain production districts (sample districts with higher area shares of grains (higher than sample median)), pulses/legumes production districts, fruits/vegetables production districts. Again, variables consisting of cumulative potassium inputs (e.g., $K_{jtc}A_{jt}$ in districts with lower grain area shares and higher fruits/vegetables area shares, as well as districts with both higher and lower pulses/legumes area shares, and $K_{jtc}N_{jt}$ in districts with higher grain area shares and lower fruits/vegetables area shares) are often selected among the most influential input variables, while variables consisting of K_{jt} are not. The importance of cumulative effects of potassium in Table 4 therefore holds generally broadly across diverse cropping systems.

4.1.1.2. Robustness of results in the presence of autocorrelation. The presence of significant autocorrelation can pose challenges to the standard LASSO estimation methods, because various methods developed for independent data are not suitable for dependent data, including autocorrelated ones (Ahrens et al., 2020; Chernozhukov et al., 2021). We therefore also conduct a robustness check in the presence of potential autocorrelation using DLD-data. Appendix B (B.5) provides more detailed descriptions of the methodology and results. As is shown in Table B4, the key implications of our primary results are robust, i.e., the cumulative value of potassium K_{jtc} consistently remains among the dominant predictors of the output.

4.1.1.3. Robustness of results based on supplementary Neural Network analysis. As was mentioned earlier, our primary analyses are confined to LASSO application to the linear characterization of production systems (i.e., translog production function). While it is more challenging to conduct similar analysis using other methods like Neural Network analysis that directly model nonlinear relations in the production system, it is still useful to conduct additional robustness checks through exploratory analyses, employing the recent development in the field to partly overcome its "black box" nature. We extend the approach by Zhang et al. (2018) which identifies an indicator of the relative importance of each input variable within the production system derived from the Neural Network analysis, using the methodology proposed by Garson (1991) and Goh (1995) that dissect the model weights to determine relative importance (due to space limitation, Appendix C provides more detail of the estimation procedures and results). Fig. C1 in Appendix C implies that cumulative quantity of potassium (K_{itc}) is still identified as one of the most important predictors of the output, which is generally consistent with our primary results based on LASSO. The exploratory results based on the Neural Network analysis indicate that our primary

Table 5
GMM-LASSO approximation of translog production function for farm household level data (VDSA-data).

LASSO-selected inputs[a]	GMM-LASSO selection based on AIC	GMM-LASSO selection based on BIC
	Coef. (std.err)	Coef. (std.err)
$\boldsymbol{K_{itc}}$	**0.214*** **	**0.228*** **
	(0.078)	**(0.066)**
$\boldsymbol{K_{itc}A_{it}}$	**−0.177** **	**−0.243*** **
	(0.088)	(0.062)
$E_{it}E_{it}$	0.527***	0.492***
	(0.096)	(0.067)
A_{it}	0.162*	0.227***
	(0.094)	(0.078)
$W_{it}W_{it}$	0.066*	0.080**
	(0.039)	(0.038)
R_{it}	0.091*	0.109**
	(0.054)	(0.051)
$E_{it}L_{it}$	0.055	
	(0.091)	
$P_{itc}P_{itc}$	−0.046	
	(0.061)	
$A_{it}N_{it}$	−0.020	
	(0.087)	
Year * Village dummies	Included	Included
Farm household fixed effects	Included	Included
Intercept	Included	Included
Samples	3041	3041
p-value (H_0: under identified)	0.006	0.000
p-value (H_0: not over identified)	0.296	0.376
Information Criteria Statistics of GMM LASSO selection as shown in Appendix A, Eq. (A4)	AIC = 34.807	BIC = 139.124

Source: Authors. ***1 % **5 % *10 %.
AIC = Akaike Information Criterion; BIC = Bayesian Information Criterion.
[a] A = land, W = capital, L = labor, E = other expenditures, R = irrigation, N = nitrogen, P = phosphate, K = potassium.

Table 6
Dynamics of fertilizer nutrient use decision at extensive margins (Arellano and Bond, 1991) dynamic panel linear probability model.

Variables	Nitrogen		Potassium	
	One step GMM	Two step GMM	One step GMM	Two step GMM
	Coef. (std. err)	Coef. (std. err)	Coef. (std. err)	Coef. (std. err)
Lagged dependent variable (whether used respective fertilizer at $t-1$)	−0.055 (0.230)	−0.241 (0.226)	−0.597** (0.249)	−0.670** (0.265)
Price at t	−0.006** (0.003)	−0.008*** (0.002)	−0.063** (0.031)	−0.069** (0.032)
Year dummies	Included	Included	Included	Included
Other variables Z_{it}	Included	Included	Included	Included
Intercept	Included	Included	Included	Included
Samples	2944	2944	2944	2944
p-value (H_0: no serial correlation at third lag)	0.740	0.985	0.297	0.371
Most recent lag of dependent variable used as an IV	$t-3$	$t-3$	$t-3$	$t-3$
Number of IVs	18	18	18	18
p-value (H_0: no overidentification)	0.356	0.356	0.397	0.397

Source: Authors. ***1 % **5 % *10 %.

Table 7
Dynamics of fertilizer nutrients use at farm household level – Wooldridge (2005) random effects dynamic panel Tobit regression (5).

Variables	Nitrogen (in 100 kg)		Potassium (in 100 kg)	
	Coef. (std. err)	Coef. (std. err)	Coef. (std. err)	Coef. (std. err)
Lagged dependent variable (quantity at $t-1$)	−0.065 (0.054)	−0.078 (0.050)	−0.556*** (0.051)	−0.558*** (0.050)
Price[a] at t	−0.445* (0.247)		−0.236 (0.646)	
Price (average between t and $t-3$)		−1.851*** (0.313)		−0.314 (0.250)
Initial period value of dependent variable	Included	Included	Included	Included
Year * Village dummies	Included	Included	Included	Included
Other variables Z_{it}	Included	Included	Included	Included
Time invariant random effects	Included	Included	Included	Included
Intercept	Included	Included	Included	Included
Samples	1172	1172	1172	1172
p-value (H_0: coefficients are jointly insignificant)	0.000	0.000	0.000	0.000

Source: Authors. ***1 % **5 % *10 %.
[a] Price of respective fertilizer nutrients.

results in Table 3 may not be simply the artifacts of LASSO-based analysis.

4.1.2. Farm household-level evidence (VDSA-data based on GMM-LASSO)

Similarly, Table 5 summarizes the results for (2) using GMM-LASSO selected variables based on AIC and BIC. As expected, GMM-LASSO based on BIC selected fewer parameters than that based on AIC, which is consistent with the application of AIC and BIC in other general contexts. Similar to the case of DLD data (Table 3 and Table 4), more variables containing K_{itc} tend to remain as influential than variables consisting of K_{it}. Furthermore, diagnostic statistics indicate that, for the models shown in Table 5, IVs provide sufficient identification power (i.e., the null hypothesis of underidentification is rejected) but also do not cause overidentification, suggesting that results are unbiased and consistent from GMM standpoint.

4.1.3. Dynamics of potassium fertilizer use

The significant cumulative effects of certain fertilizer nutrients like potassium, compared to other major nutrients like nitrogen observed in Table 4 and Table 5, can have major implications on the dynamic nature of demand for these nutrients. Table 6 and Table 7 show indicative evidence based on Eqs. (4) and (5).

4.1.3.1. Arellano and Bond (1991) dynamic panel linear probability model. Table 6 shows the results of (4) based on Arellano and Bond (1991) dynamic panel LPM. For each of nitrogen and potassium, we show both one-step GMM (which is less efficient but more robust) and two-step GMM results (which is more efficient under the assumption that estimated weighting matrix is correct). The rows for Arellano & Bond autocorrelation tests indicate the absence of autocorrelation with the 3rd lag. We therefore use $Q_{i,t-3}$ and earlier lags as additional IVs. The bottom row indicates that using all lags of Q_{it} up to $Q_{i,t-3}$ satisfies the orthogonality conditions of all IVs.

Results indicate that the likelihood of using potassium in t is significantly negatively affected by that in $t-1$. This is consistent with the findings in Table 4 and Table 5. It is the aggregate quantity of potassium in t and $t-1$ that matters for the production, which implies that, conditional on other exogenous shocks, quantities in t and $t-1$ substitute each other. Importantly, these results do not hold for nitrogen. The contrasting results for nitrogen are again consistent with the findings in Table 4 and Table 5 in which cumulative effects are generally absent for nitrogen, unlike potassium.

4.1.3.2. Wooldridge (2005) panel dynamic Tobit model. Table 7 summarizes the results of Wooldridge's (2005) panel dynamic Tobit Eq. (5). In order to highlight the implications on the demand response to price changes, in particular its lagged nature, we also estimate additional models with average price during the last 4 years (between t and $t-3$).

The quantity of potassium used in current year (t) is statistically significantly negatively affected by the quantity used in the previous year ($t-1$). Such negative relations indicate that potassium in the previous year and current year are partly substitutes, consistent with aforementioned findings of significant cumulative effects of potassium. Such dynamic relations are statistically insignificant for nitrogen, which is again consistent with the aforementioned findings of the absence of cumulative effects for nitrogen.

4.2. Simulation results

4.2.1. Results for the productivity effects (6)

Table 8 through Table 10 summarize the simulation results of (6), i.e., how significantly, in terms of magnitude, the production technology in the current year is affected by the potassium use in the previous year, and (7), i.e., how the actual production is affected when the dynamic responses in potassium use in the current year is also considered. It is important to note that the increase in production in $t+1$ should be

Table 8
Productivity increases (%) in t due to cumulative effects of a 10 % increase of potassium in $t-1$ (for all samples; corresponding to Table 4 production function results).

Corresponding LASSO results in Table 4	Sample minimum	Sample average	Sample maximum
$\lambda = 690.65$ (RLASSO-selected)	0.15	0.63	0.89
$\lambda = 600$	−0.25	0.91	1.47
$\lambda = 500$	−0.79	0.90	1.63
$\lambda = 400$	−0.79	0.94	1.76

Source: Authors.

interpreted more as representing a carryover of production from t to $t+1$, rather than a pure net increase in $t+1$. This is because a 10 % increase in potassium use in t (K_{jt}) translates into less than a 10 % increase in the actual quantity of potassium affecting the production in t (K_{jtc}) (since K_{jtc} partly depends on $K_{j,t-1}$), and thus lower production in t than otherwise (i.e. if the increase in K_{jt} is fully used up in production in t). Nonetheless, the magnitude of the carryover of productivity from t to $t+1$ is an important indicator of the significance of the cumulative effects of potassium.

Table 8 summarizes the simulated results for all district samples based on production function estimates from Table 4. Results based on the RLASSO-selected model indicate that, for a typical district, a 10 % increase in potassium use in year t has the effect of increasing the productivity in $t+1$ (or carrying over the productivity from t to $t+1$) by 0.63 % compared to the baseline scenario in which no such shock occurred, even if the amount of inputs including potassium in $t+1$ remains the same as pre-shock period. The effects are also heterogeneous, ranging from 0.15 % at the sample minimum to 0.89 % at the sample maximum.

The results under less parsimonious production function formulas (estimated with lower λ in Table 4) are of similar magnitude to the RLASSO-selected results, albeit with generally larger distributions across district samples. These results may reflect that RLASSO-selected results may provide more precise estimates than other models thanks to reduced multicollinearity.

Table 9 and Table 10 summarize the similar results for sub-sample districts defined by soil attributes and dominant cropping systems, which correspond to production function estimates in Tables B2 and B3. These tables indicate a certain degree of heterogeneity across these subsamples, in addition to heterogeneity within each subsample. While it is beyond the scope of this paper to examine how exactly the dominant soil types of cropping systems interact with the cumulative effects of potassium, the results generally indicate larger overall effects in districts with higher soil organic contents, higher soil alkalinity, lower soil drainage, and districts with fruits/vegetable production shares. In particular, the productivity effect in the district with lower soil drainage is about 4 times larger on average than those in other districts (1.68 % vs., 0.40 %).

4.2.2. Results for the multi-year production effects (7)

Fig. 4 through Fig. 8 illustrate various key aspects of the medium-term effects of a shock in potassium use described in subsection 3.2.2. Specifically, we focus on (a) overall average effects, (b) accuracy of results based on RLASSO-selected estimates in Table 4, (c) the extent of heterogeneity in the production effects trajectory with respect to the land area of districts (A_{jt}), which is, as was discussed above, found to exhibit significant interactive effects with potassium, (d) cumulative effects of permanent reduction of potassium, which leads to the shock equivalent to 10 % increase in potassium use, and (e) the extent of heterogeneity in the effects across average soil drainage properties of the district, which is found to lead to the maximum heterogeneity based on the results in subsection 3.2.2. To be consistent with the results in the previous subsection, for (a)–(e), we consider the case in which there was a shock of a 10 % increase in potassium in t.

Table 9
Productivity increase (%) in t due to cumulative effects of a 10 % increase of potassium in $t-1$ (corresponding to Table B2 results).

Soil attributes	Sub-samples districts based on soil attributes	Sample minimum	Sample average	Sample maximum
Organic contents	High	0.25	1.18	1.60
	Low	0.19	0.84	1.20
Alkalinity (pH)	High	0.71	1.22	1.56
	Low	0.00	0.00	0.00
Drainage	More excessive	0.09	0.40	0.60
	Poorer	0.94	1.68	2.17

Source: Authors.

Table 10
Productivity increase (%) in t due to cumulative effects of a 10 % increase of potassium in $t-1$ (corresponding to Table B3 results).

Cropping systems	Sub samples with higher area shares of the relevant crop group	Sample minimum	Sample average	Sample maximum
Grains	Higher	0.00	0.80	1.02
	Lower	0.18	0.89	1.20
Pulses	Higher	0.18	0.89	1.20
	Lower	0.22	0.92	1.24
Fruits/Vegs	Higher	0.22	1.01	1.40
	Lower	0.00	0.78	0.96

Source: Authors.

Fig. 4 illustrates the result for (a), together with 95 % confidence intervals estimated through a bootstrap method by Efron and Tibshirani (1993). Fig. 4 indicates that a 10 % increase in potassium in $t=1$ leads to approximately 0.9 % increase in total production outputs in that year. Second, and more importantly, the aforementioned shock has lingering, albeit diminishing, effects over several subsequent years. In year $t=2$, the production effect relative to the baseline declines to about 0.3 %, despite the cumulative effect of potassium from $t=1$, mainly because the increase in potassium in $t=1$ significantly reduces potassium in $t=2$ (as shown in Table 7). The decline in potassium use in $t=2$ then has further cumulative effects into $t=3$, at which point the production actually drops 0.2 % below the baseline production level, even though potassium use in $t=3$ somewhat reverses upward (in response to lower potassium use in $t=2$). This upward reversal in potash use in $t=3$ further reverses the production back up to 0.13 % above the baseline production level. These partially lagged responses continue beyond $t=4$. However, over time, the production gradually converges back to the baseline production levels.

Fig. 5 illustrates the results for (b), i.e., how this estimated trajectory of production effects may differ if we use results from less parsimonious production function forms (results with lower λ in Table 4). As illustrated in Fig. 5, it turns out that estimated trajectories are generally similar when results from parsimonious production function form (the case of $\lambda=400$ in Table 4) is used (results for the case of $\lambda=500$ or $\lambda=600$ in Table 4 are virtually identical to that of $\lambda=400$ and therefore not shown in Fig. 5. These patterns suggest that RLASSO-selected production function estimates are reliable approximations of true production function, and underscore the critical importance of the cumulative effects of potassium.

Fig. 6 illustrates the results for (c), i.e., how the estimated trajectory in Fig. 4 (which is a sample average) varies across districts with different total land area (A_{jt}) which our estimated production functions generally indicate as significant factors affecting the marginal effects of potassium. In doing so, we trace the trajectories for districts with maximum, 75th, 50th, and 25th percentiles, as well as minimum land areas. As is shown in Fig. 6, production effects (both upward and downward swings over time) are more pronounced in districts with larger land areas than in districts with smaller land areas. In the district with the largest land area, the effects are generally about 40 % larger than those with median land areas, while they are generally only about 1/5 in magnitude in the district with the smallest land area. The effects in districts with 75 percentile of land areas are about 10 % and 20 % larger than the effects in districts with 50 and 25 percentiles of land areas.

Fig. 7 illustrates the results for (d), i.e., how the production effects in each year in Fig. 4 accumulate over time. Similar to Fig. 4, 95 % confidence intervals estimated through a bootstrap method are also included. Fig. 7 may also be considered as a scenario in which the shock that increased potassium use in $t=1$ (such as a significant reduction in potassium price through subsidies or other price control measures)

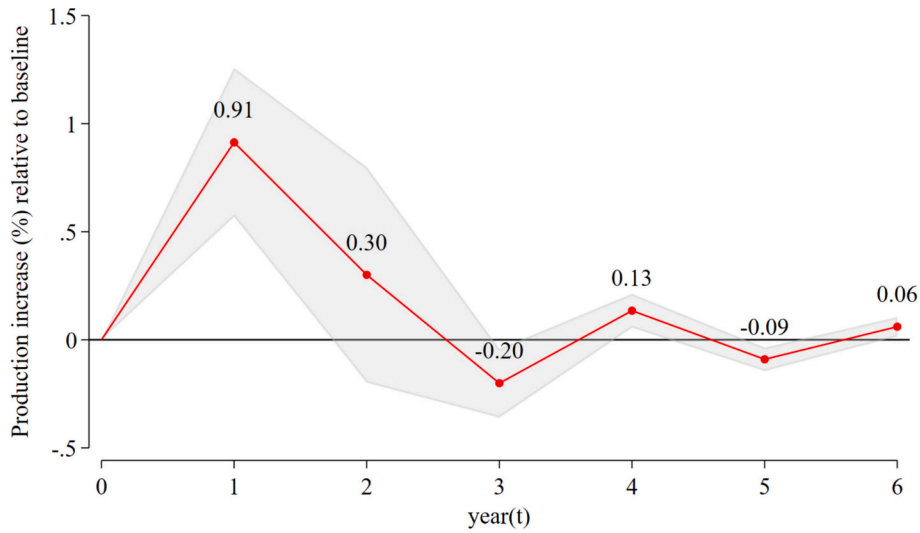

Fig. 4. Trajectory of production effects of potassium shock in $t = 1$ (average among DLD-data district samples)[a, b].
Source: Authors.
[a]The shaded area represents the 95 % confidence intervals estimated through a bootstrap method by Efron and Tibshirani (1993).
[b]The simulation in this figure is based on the following configurations:
(i) Shock magnitude: a 10 % increase in potassium in $t = 1$.
(ii) Unit: Production increase in % relative to the baseline scenario of no shock.
(iii) Sample: 496 districts in each year.
(iv) Estimator: Eq. (7).
(v) Parameter sources: Estimated production function coefficients in Table 4 (under the Post Double-Selection LASSO) and potassium use function coefficients in Table 7.

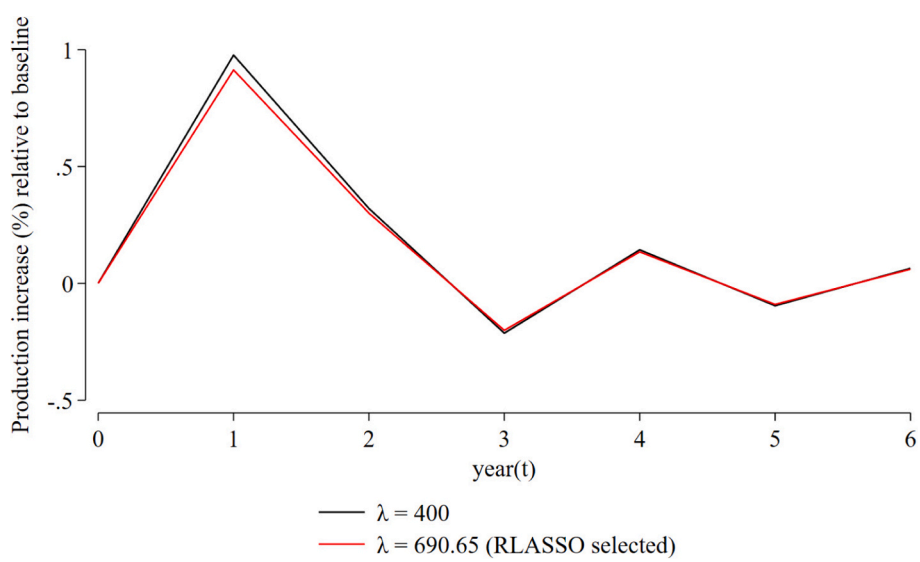

Fig. 5. Same results for Fig. 2 based on different Table 4 production function estimates[a, b].
Source: Authors.
[a]Note: "λ" is the value of corresponding LASSO penalty weight.
[b]The simulation in this figure is based on the following configurations:
(i) Shock magnitude: a 10 % increase in potassium in $t = 1$.
(ii) Unit: Production increase in % relative to the baseline scenario of no shock.
(iii) Sample: 496 districts in each year.
(iv) Estimator: Eq. (7).
(v) Parameter sources: Estimated production function coefficients in Table 4 and potassium use function coefficients in Table 7.

remains permanent from $t = 2$ on, while Fig. 4 refers to the scenario in which the shock only occurs in $t = 1$. Fig. 7 suggests that these permanent shocks lead to the production effects converging around 1.1 % above the baseline production level. Such convergence, despite shocks being permanent, may be due to the significant dynamics in potassium use across years, in which potassium use responds significantly negatively to the increase in the previous year even if lower potassium price stimulates increased potash uses.

Lastly, Fig. 8 shows the results for (e), which provides an illustrative example of how the trajectory of production effects in Fig. 4 varies in

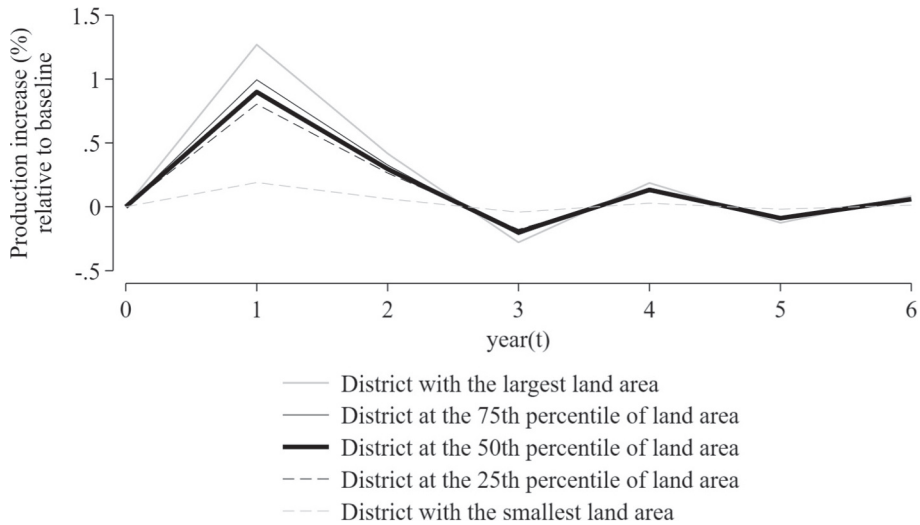

Fig. 6. Heterogeneity in Fig. 3 results across district land-area quartiles[a].
Source: Authors.
[a]The simulation in this figure is based on the following configurations:
(i) Shock magnitude: a 10 % increase in potassium in $t = 1$.
(ii) Unit: Production increase in % relative to the baseline scenario of no shock.
(iii) Sample: 496 districts in each year.
(iv) Estimator: Eq. (7).
(v) Parameter sources: Estimated production function coefficients in Table 4 (under the Post Double-Selection LASSO) evaluated at respective sample percentiles of land area and potassium use function coefficients in Table 7.

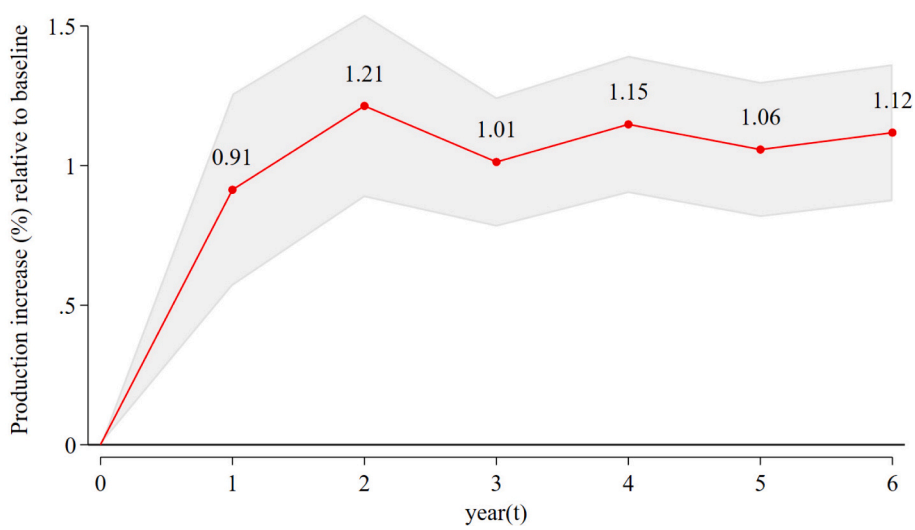

Fig. 7. Cumulative effects of 10 % shock in potassium use in t on production during subsequent years relative to the baseline[a, b].
Source: Authors.
[a]The shaded area represents the 95 % confidence intervals estimated through a bootstrap method by Efron and Tibshirani (1993).
[b]The simulation in this figure is based on the following configurations:
(i) Shock magnitude: a 10 % increase in potassium in $t = 1$.
(ii) Unit: Production increase in % relative to the baseline scenario of no shock.
(iii) Sample: 496 districts in each year.
(iv) Estimator: Eq. (7) with accumulation of effects at each year (t).
(v) Parameter sources: Estimated production function coefficients in Table 4 (under the Post Double-Selection LASSO) evaluated at respective sample percentiles of land area and potassium use function coefficients in Table 7.

magnitude across subsamples of districts with different attributes other than land areas. Fig. 4 specifically illustrates the heterogeneity with respect to soil drainage properties of the districts, which appear to lead to one of the most pronounced heterogeneity (based on Table 8). Fig. 8 suggests that, similar to the heterogeneity with respect to land areas in Fig. 6, the variations in soil drainage properties generally lead to proportional changes in the extent of production effects. Specifically, in districts with poor soil drainage, the production effects are approximately 85 % larger than those in average districts and about 4 times larger than those in districts with excessive soil drainage. Although not

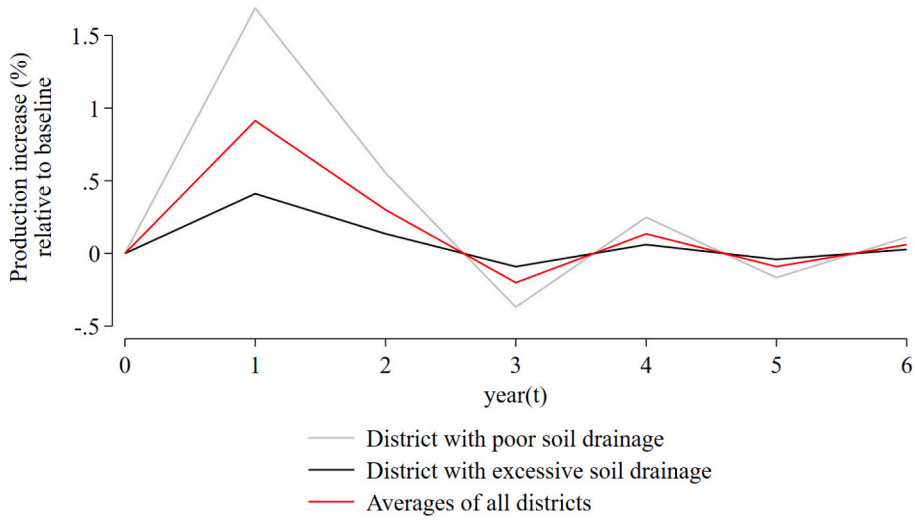

Fig. 8. Heterogeneity in Fig. 4 results across districts with different soil drainage properties[a].
Source: Authors.
[a]The simulation in this figure is based on the following configurations:
(i) Shock magnitude: a 10 % increase in potassium in $t = 1$
(ii) Unit: Production increase in % relative to the baseline scenario of no shock
(iii) Sample: 496 districts in each year (248 districts with high soil drainage and 248 districts with low soil drainage)
(iv) Estimator: Eq. (7) with accumulation of effects at each year (t)
(v) Parameter sources: Estimated production function coefficients in Table B2 (under the Post Double-Selection LASSO) evaluated at respective sample percentiles of land area and potassium use function coefficients in Table 7

shown due to space limitation, the patterns of heterogeneity are generally similar when we focus on other dimensions of soil attributes or cropping systems (based on sub-sample production function estimates in Tables B2 and B3). These patterns suggest that the overall average trajectory of production effects in Fig. 4 offers informative insights into the relative magnitudes of the repercussions and speed of convergence back to the baseline level (in subsequent years) following the initial shock in potassium use.

5. Discussions

The results presented in the previous sections have important implications across various aspects of intertemporal linkages within dynamic production systems. Overall, these production systems remain complex. However, capturing the key elements of underlying production function and the essential nature of intertemporal linkages embedded in it offers us a way to understand better the essential driver with which these dynamic production systems operate. In our example, potassium fertilizer, combined with other supplementary inputs, is identified as one of such critical drivers within the dynamic production system in India. This section provides more granular discussions of the implications of these findings.

5.1. Approximation of complex production function beyond yield determinants

Many previous agronomic studies using the LASSO-approach mainly focused on analyzing the determinants of yield (Asadollah et al., 2024; De Clercq and Mahdi, 2024; Sunoj et al., 2024). While focusing on yields allows potentially more delicate analyses of yield determinants, applying LASSO to the flexible translog production function, as in our case, offers unique but also supplementary insights to previous studies using LASSO. It is particularly so in the contexts where the change in land use itself from year to year can be significant due to factors including worker availability, which can vary in different years, among others. Similarly, extending the LASSO beyond yield is important when there are complex interactive effects between different production factors (either complementary or substituting effects). Our results reemphasize this. For example, our LASSO approach efficiently identified that an increase in overall productivity resulting from a shock (a 10 % increase) in previous year's potassium use can vary considerably depending on the land size in the current year (among all the inputs used as well as the year of their uses), ranging from 0.15 % for the smallest farm to 0.89 % for the largest farm (Table 8). Such a finding constitutes a significant departure from conventional focus on yield, for which the heterogeneous effects with respect to land size are less frequently considered.

5.2. Cumulative effects of potassium in nonexperimental settings

Our results suggest that potassium plays significant roles, particularly through its cumulative effects, on agricultural production at field levels under farmers' actual production practices. Our results complement rich evidence in experimental agronomic settings. They shed light on the relative importance of different agricultural inputs and their potential cumulative effects in the nonexperimental setting that may reflect production environments in which average farmers operate in reality, given their specific agroecological conditions and resource endowments. Importantly, by aggregating across multiple crops, our analyses also offer a more representative picture of the agronomic characteristics in India and at the farm household level. Our findings have important implications not only at single crop levels but also for the overall agricultural sector and all crop production activities for the household. It is also important to note that the cumulative effects of potassium broadly hold across different levels of the use of other inputs. As shown, significant cumulative effects of potassium are identified within the translog production function framework that explicitly accounts for the use of other production factors, as well as the complex interactive effects among these other production factors on output. This suggests that the identified cumulative effects of potassium are less likely due to the failure of the model to correctly separate out the effects from the other inputs that happen to be highly correlated with potassium

use, which can sometimes be difficult to differentiate under nonexperimental settings. Specifically, while Harper et al. (2012), one of the few studies on potassium carryover in the literature, showed that 70 % of potassium K was carried over to the next production season under experimental settings in the US (Harper et al., 2012), our findings extend related evidence to a non-experimental setting in India, showing that a 10 % increase in potassium fertilizer increases the overall productivity in the following year by 0.63 %. Furthermore, the fact that the cumulative effects of potassium remain significant also in the parsimonious representation of production function by LASSO suggests that the cumulative effects of potassium are sizeable with substantial implications on overall production, even compared to the other production inputs.

5.3. Consistent importance of the cumulative effects of potassium across agroecology and cropping systems despite heterogeneity

Results also suggests that the relative importance of potassium as well as its cumulative effects in crop production may hold relatively consistently under diverse agroecology (based on soil characteristics) and cropping systems within India. In other words, earlier agronomic results that have suggested such cumulative effects of potassium are potentially relevant across wider geographic regions than previously thought.

At the same time, the set of other inputs found among LASSO-selected potassium variables still varies across these ecologies and cropping systems, suggesting that the marginal effects of potassium can still vary depending on the use of other inputs. For example, significantly positive coefficients for $\boldsymbol{K_{jtc}A_{jt}}$ in the most parsimonious models in Table 5 suggest that, nationwide, the output elasticity with respect to cumulative quantity of potassium used is larger (and positive) if more land is used in production. However, within the specific samples studied in VDSA-data (Table 6 and Table 7), the signs of coefficients for $\boldsymbol{K_{itc}A_{it}}$ are significantly negative, suggesting that the output elasticity with respect to cumulative quantity of potassium used is larger (and positive) if less land is used in production. These differences suggest that the joint effects of cumulative quantity of potassium and land can be significantly heterogeneous at local levels. Similarly, in Table B3, the output elasticity of cumulative quantity of potassium varies more in response to the size of the land used in areas with higher shares under fruits/vegetables and lower shares under grains (as $\boldsymbol{K_{jtc}A_{jt}}$ is selected by LASSO as the key predictor of output). However, the same elasticity varies more in response to nitrogen quantity applied in areas with more grains and less fruits/vegetables (as $K_{jtc}N_{jt}$ is selected by LASSO). These differences suggest that the joint effects of cumulative quantity of potassium and certain other inputs like land or nitrogen can be significantly heterogeneous across cropping systems.

Importantly, however, regardless of the directions of these interactive effects, outputs remain more responsive to cumulative quantity of potassium, rather than current year's application of potassium alone. Such consistency in the relative importance of cumulative quantity of potassium (compared to current year's alone) across broad production environment within India is the central significance of our findings.

The magnitude of the estimated cumulative effects, derived from a simple simulation exercise, is also noteworthy. As mentioned above, the implication of the cumulative effects of potassium is the carryover of productivity between t to $t+1$, which is different from genuine growth in productivity. Nonetheless, given that the annual growth rate of agricultural productivity in developing countries like India is typically in the order of 1 % (USDA, 2025), the observed cumulative effects of potassium (in which a 10 % increase in its use in the previous year has the effect of carrying over the productivity equivalent to 0.63 % to the current year) can be considered sizable.

5.4. Implications of dynamic patterns of potassium use by farmers

Our analyses of the dynamic patterns of potassium application by farmers further suggest that the identified cumulative effects of potassium are also causing the current year's potassium application to depend significantly on the previous year's application of potassium. The relative importance of potassium and its cumulative effects is significant enough to the extent that it also affects farmers' actual potassium use in dynamic ways. Farmers may either be informed of such cumulative effects from extension services or their own experiences.

This pattern offers insights into how agronomic technological characteristics translate into the actual agricultural outputs. Specifically, compared to the scenarios *without* farmers' adjustments in potassium use, agricultural production in India is more stabilized between the current year and the prior year. This is because of the following mechanisms: cumulative potassium use for the previous year and the current year overall has positive effects on outputs; farmers who used more potassium in the prior year use less potassium in the current year, leading to more stable cumulative potassium inputs and thus more stable outputs across years.

Nonetheless, the simulation exercise of the trajectory of the effects over multiple years reveals that such dynamic relations in potassium use across years, combined with cumulative effects of potassium, lead to complex repercussions of a shock in potassium over the subsequent years. Generally, as shown in Section 4, such repercussions can persist, albeit in a diminishing manner, for several years, and their effects can remain substantial, particularly in districts with certain characteristics, such as a larger land area and poorer soil drainage. Specifically, our results in Fig. 4 suggest that combined effects of years 2 and afterward, 0.78 % (which is the sum of absolute values of deviations from the baseline scenario, i.e., $0.78 = 0.30 + 0.20 + 0.13 + 0.09 + 0.06$), are in a similar order of magnitude as the effect of 0.91 % in year 1. The evidence of such significant persistent effects is a significant departure from the past studies of fertilizer in South Asia (e.g., Takeshima, 2017), which typically did not consider such medium-term effects of fertilizer price shocks. Our production function analyses suggest that cumulative effects fade two years after the initial shock in potassium use when combined with significant negative dynamics in potassium use. This implies that an increase in potassium can also lead to even lower production 2 years later, than the production under the baseline scenario in which no such shocks occur. Of course, agricultural production can also be affected by other factors. However, the crucial role that the cumulative quantity of potassium plays in production (which we find in our production function analyses) suggests that such repercussion of a shock in potassium use in earlier years remains one of the dominant determinants of agricultural production and its fluctuations in subsequent years.

5.5. Policy implications for potassium toward agricultural production growth in India

Aside from cumulative effects, the generally significant predicting power of potassium in production function, relative to other inputs, is consistent with the hypothesis that relatively low potassium use (compared to nitrogen and phosphorus) in India, as described in the background section, is one of the key constraints in agricultural production in the country.

Our results have important policy implications. First, promoting increased potassium use may be beneficial in increasing agricultural production in India, given the significant roles of potassium in production, as well as the relatively low current potassium application in India compared to other nutrients and the rest of the world. Second, in terms of developing improved fertilizer dosage recommendations for farmers

(which remains an essential aspect of fertilizer policies in South Asia (e.g., Beg et al., 2024)), the potential cumulative effects of potassium should be more carefully examined at specific localities and should be incorporated in such recommendations. In doing so, monitoring and keeping a record of previous years' potassium use (as in DLD-data and perhaps even at lower administrative levels) remains critical, as the recommended dosage can depend on the use in the previous years. Third and lastly, in terms of price policies, price stabilization efforts by the Indian government may be more justified for inputs like potassium fertilizer. This is because, as the demand for potassium is inelastic to price (compared to nitrogen fertilizer) in India, its price fluctuations can lead to higher farm income fluctuation, while inelastic demand can also cause more significant price fluctuations locally.

5.6. Limitations of the current study

While our results provide significant contributions to the literature, there are still areas for improvement that can be addressed in future studies. One of the limitations of the current study is that, due to the lack of information, we are unable to conduct more in-depth analyses of the potential mechanisms underlying the observed roles of potassium and its cumulative effects in production. For example, future studies can deepen the analyses by investigating the effects on intermediate outcomes, such as the soil fertility level (e.g., the residual level of potassium), based on additional soil data. In addition, while our results are robust for the period up to 2016 for which suitable data were available, updated analyses are needed through new sets of data to gain insights into more recent characteristics of the Indian agricultural production systems.

6. Conclusions

The cumulative effects of inputs on production, including the effects of inputs in one year on the production in the subsequent years, have been widely recognized. Nonetheless, unlike sectors outside the agriculture (e.g., cumulative effects of education on human capital formation) or the case of public goods inputs within the agricultural sector, the evidence for the private goods inputs in the agricultural sector has remained generally anecdotal or limited to agronomic trial setting, and direct field level evidence has remained scarce. Addressing this knowledge gap is important because, for example, potentially cumulative effects of inputs can affect optimal production advice conveyed by agricultural extension efforts in a particular year, or the demand for these inputs in a particular year and thus the effectiveness of policies like subsidies. Addressing this knowledge gap has been challenging, particularly within the production function framework, due to the general scarcity of long panel data on agricultural production, significant multicollinearity among inputs, and their quantity across years.

We attempted to narrow this knowledge gap through the application of LASSO and GMM-LASSO methods to the estimation of flexible translog production function using unique datasets of district-level long panel data (DLD-data) and farm-household panel data (VDSA-data) in India. Our results show that certain inputs, particularly potassium, remain one of the particularly persistent key predictors of production level, even when compared with other major inputs like land, labor, capital, irrigation, and other fertilizer nutrients. More importantly, the aggregate quantity of potassium from the current year, as well as the previous year, is a consistently more important determinant of production than the quantity of potassium from the current year alone, demonstrating the significant cumulative effects of potassium. These patterns do *not* hold for nitrogen and most other inputs. These results hold for both district level (DLD-data) and farm-household level (VDSA-data). The dynamic panel linear probability model and the dynamic panel Tobit analyses further suggest that the use of potassium used in the current year is significantly negatively affected by the quantity used in the previous year, while no such dynamics were observed for nitrogen, consistent with production function results. Simulation exercises based on the regression results on production function and dynamics of inputs use reveal that a shock represented by an initial increase in potassium fertilizer use has a relatively sizable carryover effects on the system productivity in the following year and, combined with inter-annual dynamic relation in potassium fertilizer use, leads to considerable inter-annual fluctuations in production over the course of several years during the subsequent period.

CRediT authorship contribution statement

Hiroyuki Takeshima: Writing – original draft, Visualization, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Avinash Kishore:** Writing – review & editing, Validation, Supervision, Funding acquisition.

Declaration of competing interest

Authors have no conflicts of interests with the manuscript.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.agsy.2025.104555.

Data availability

Data will be made available on request.

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